

ways to make it admissible



$h(n) \leq h^*(n)$ — Underestimation

$h(n) \geq h^*(n)$ — Overestimation

$h(n)$ — estimated value

$h^*(n)$ — actual value

$$g(A) = 200$$

$$g(B) = 200$$

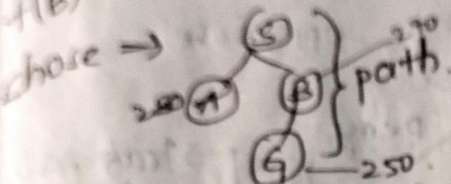
$$f(n) = g(n) + h(n)$$

Case I: Overestimation

$$\begin{matrix} h(A) = 80 \\ h(B) = 70 \end{matrix} \Bigg] > h^*$$

$$f(A) = 200 + 80 = 280$$

$$f(B) = 200 + 70 = 270 \checkmark$$



$$f(G) = g(G) + h(G)$$

$$= 200 + 50 + 0 = 250$$

No optimal ans

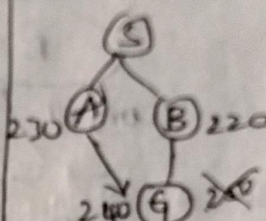
Case II: Underestimation

$$h(A) = 30$$

$$h(B) = 20$$

$$f(A) = 200 + 30 = 230$$

$$f(B) = 200 + 20 = 220 \checkmark$$



$$f(G) = 200 + 50 + 0 = 250$$

But here there is $f(A) = 230$
($230 < 250$) $\rightarrow$ explore A

$$f(G) = 200 + 40 + 0 = 240$$

$\therefore$ path $\Rightarrow$ S - A - G $\rightarrow$ optimal Ans

$\therefore$ In Underestimation, we always find optimal soln &
In overestimation we can't find optimal soln//

* Adversarial Search (Game playing)

- $\rightarrow$ a players game
 - Tic-Tac-Toe
 - chess

$\rightarrow$ Adversary — opponent $\Rightarrow$ based on opp, we need to take dec'n & move.

& opp will always try to defeat us.

Ex 4 & Friends are there & police caught them

A \ B	B	
	confess	Deny
confess	-2 -2	-3 0
Deny	0 -3	-1 -1

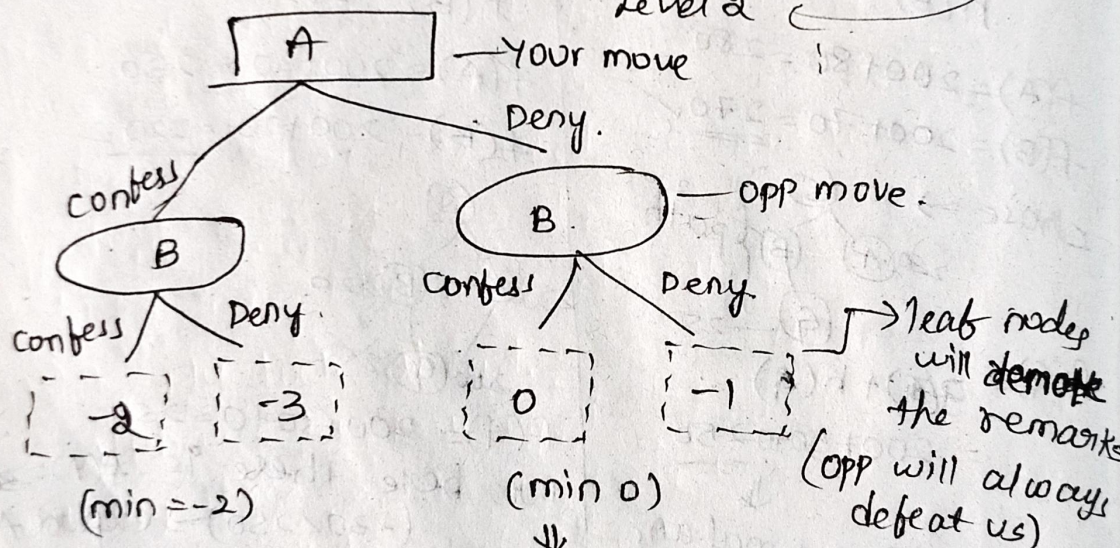
There will be remarks.
(i.e Jail)
-3 $\Rightarrow$ 3 yrs of Jail.

- Rational decision.
- Best outcome / reward

* we will draw a Game tree.

Game Tree.

Level 0 - Your move
Level 1 - opp move.
Level 2



$A \Rightarrow$ (Deny) is better option.

* Each level represent a diff layer

* leaf node represent reward

Based on reward we will go for decision making

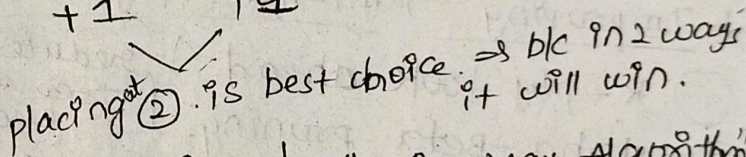
Ex 1 Tic-Tac-Toe.

0	0	X
X	①	0
②	③	X

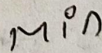
Next move is by X.

$\Rightarrow$ There are 3 possibilities

Reward $\rightarrow$ win +1
lose -1
Draw 0.



max

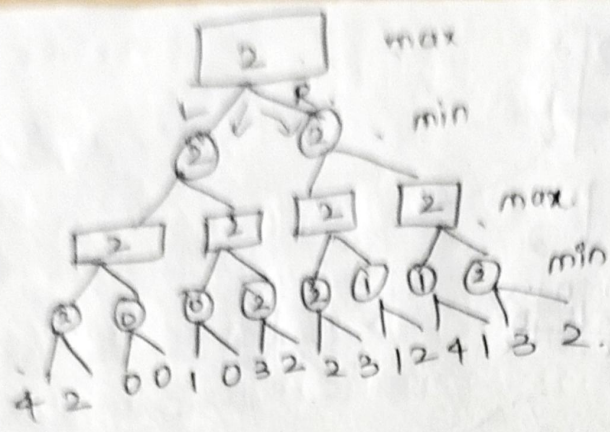


Max

Reward.

- * In min level we try to return min value from children
- * In max level we try to return max value from children
- * The branch from which max value is coming, then we go with that branch.

4/4



□ max
○ min

→ Any branch we will
→ The max we will get is 2 or less.
(2, 1, 1, 0).
→ b/c our opp will go with min value

* The probm is size of Tree.

* If there is empty tic-tac-toe, then there are many children.

* If we have to start from empty, it is not possible to evaluate the all nodes.

* The 2 ideas are used to improve this.
1. Just look at few steps and take dec?

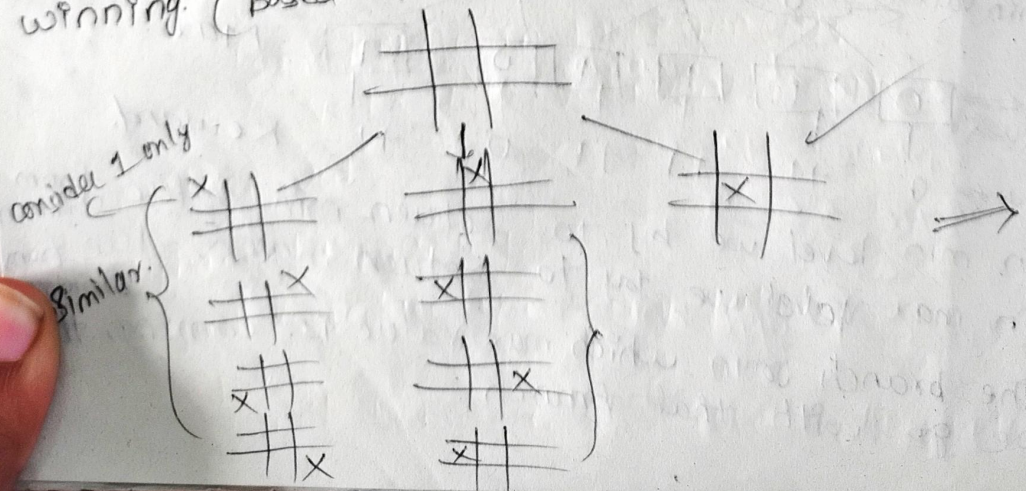
* Improving Min-max Algorithm

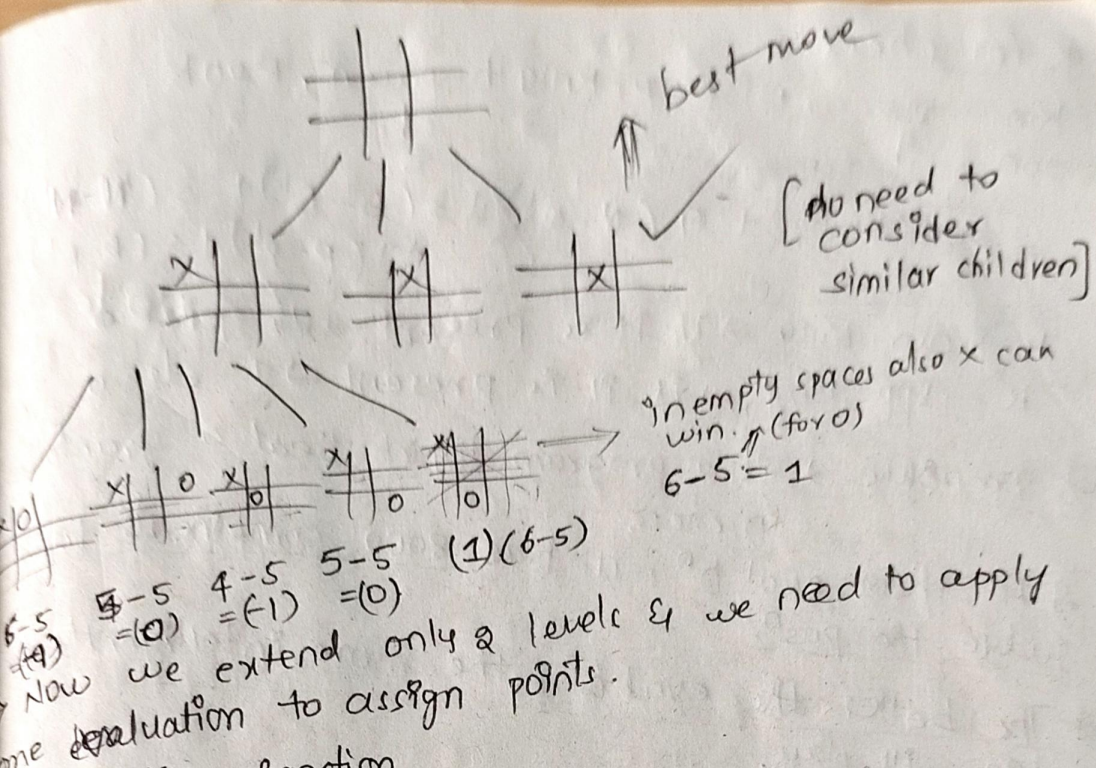
① K-ply look ahead. } reduce the size of the game tree that we need to evaluate.

② Alpha-Beta pruning.
→ using these 2, we are going to reduce the tree we are going to evaluate.

① K-ply look ahead

→ Instead of going to end, we can extend to some levels & give rewards based on the strength towards winning. (based on evaluation fⁿ).



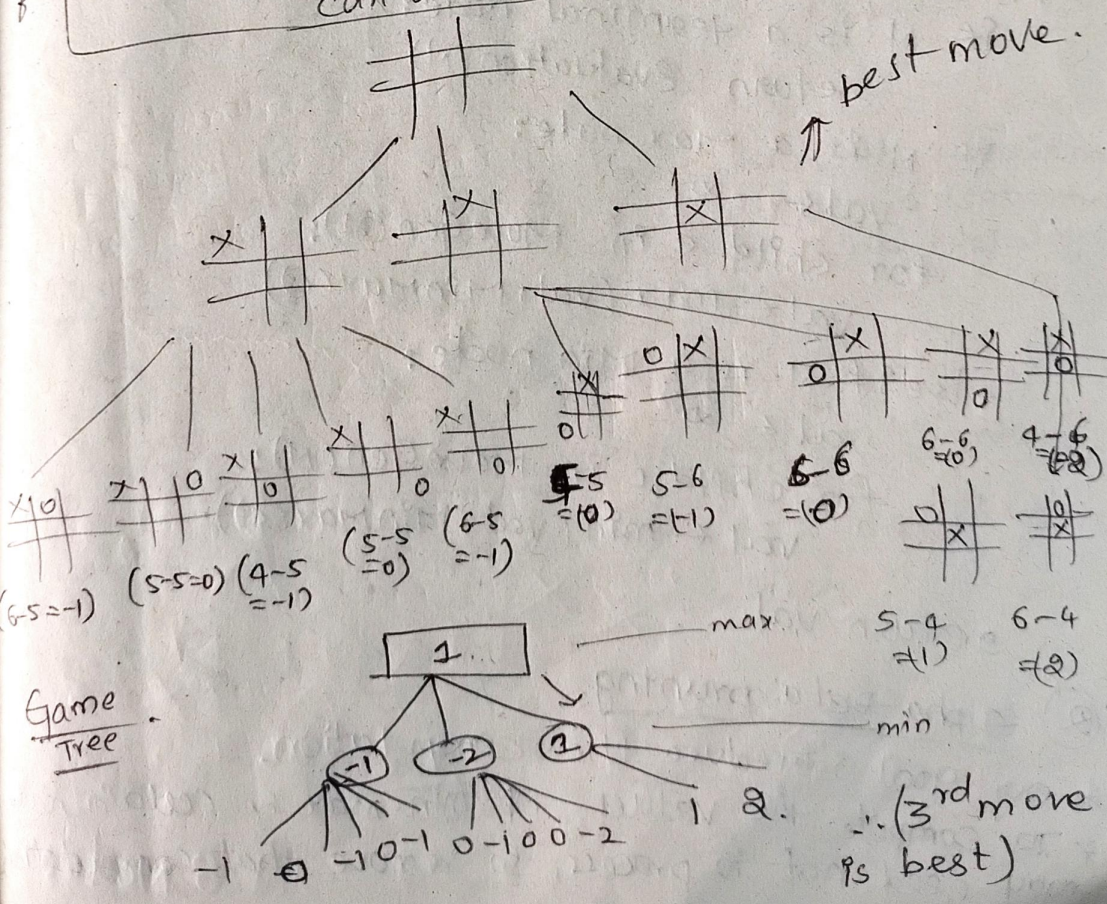


Now we extend only 2 levels & we need to apply evaluation to assign points.

Evaluation function

tic-tac-toe

eval for = No. of rows, columns, diagonals where x can win - No of rows, columns, diagonals where o can win.



* for chess:

King, Queen, Bishop, Knight, pawn, Rook.

200

9 3 3 1 5

$$= 200(K-K') + 9(Q-Q') + 3(B-B') + 3(N-N') + 1(P-P') + 5(R-R')$$

$K \rightarrow$ our King $\Rightarrow$ if it is present $\Rightarrow 1$, else 0.

$K' \rightarrow$ opp King $\Rightarrow$ if it is present $\Rightarrow 1$ else 0

* DeepBlue $\Rightarrow$ is a program which beat a grandmaster in chess.

* In their eval b^n , it has 8000 components to judge the posⁿ.

* The better the eval b^n , the more intelligently the algo will play.

Algorithm of MinMax:

MinMax(N):

if N is a terminal node:
return Evaluation(N)

if N is a Max node:
val $\leftarrow -\infty$.

for child c in MoveGen(N):
val $\leftarrow \max(\text{val}, \text{MinMax}(c))$

elseif N is a Min node:
val $\leftarrow \infty$

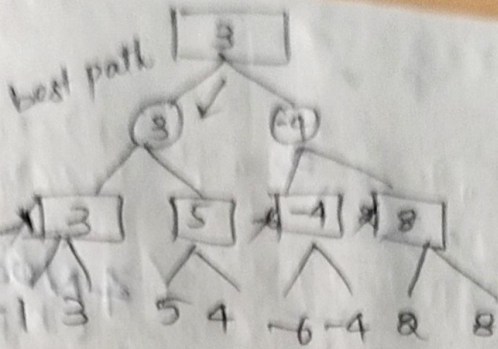
for child c in MoveGen(N):
val $\leftarrow \min(\text{val}, \text{MinMax}(c))$.

return val.

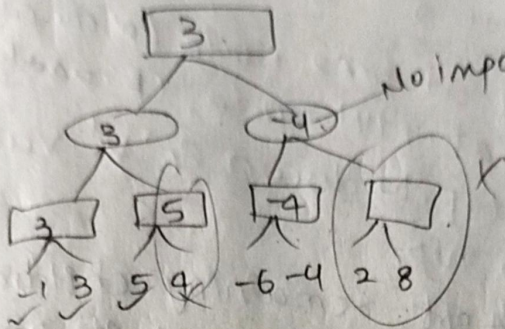
② Alpha-Beta pruning.

* Our goal is to reduce the computation.

* To compute the values of min-max of certain nodes may need not to process, so remove that computation.

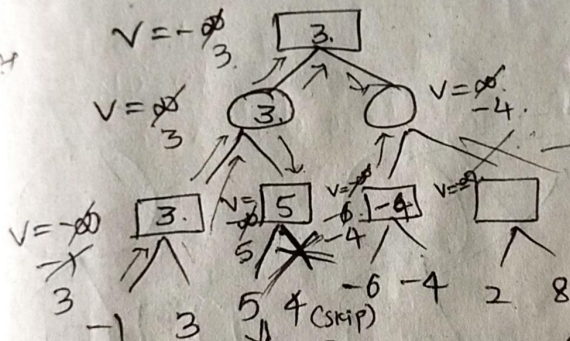


Total no of nodes we visit = $8 + 9 + 2 + 1 = 15$



we are cutting down the unaffected ones c/s "Pruning"

when we go to 4, the node 3 does not have any impact b/c $5 > 3$ so, no need to go to 4. like this we can reduce computations.



stop b/c it is not going to impact the Parent

after 5, $[3 < 5] \Rightarrow$ so the remaining children doesn't matter b/c parent didn't get any impact.

2 cases of cut off

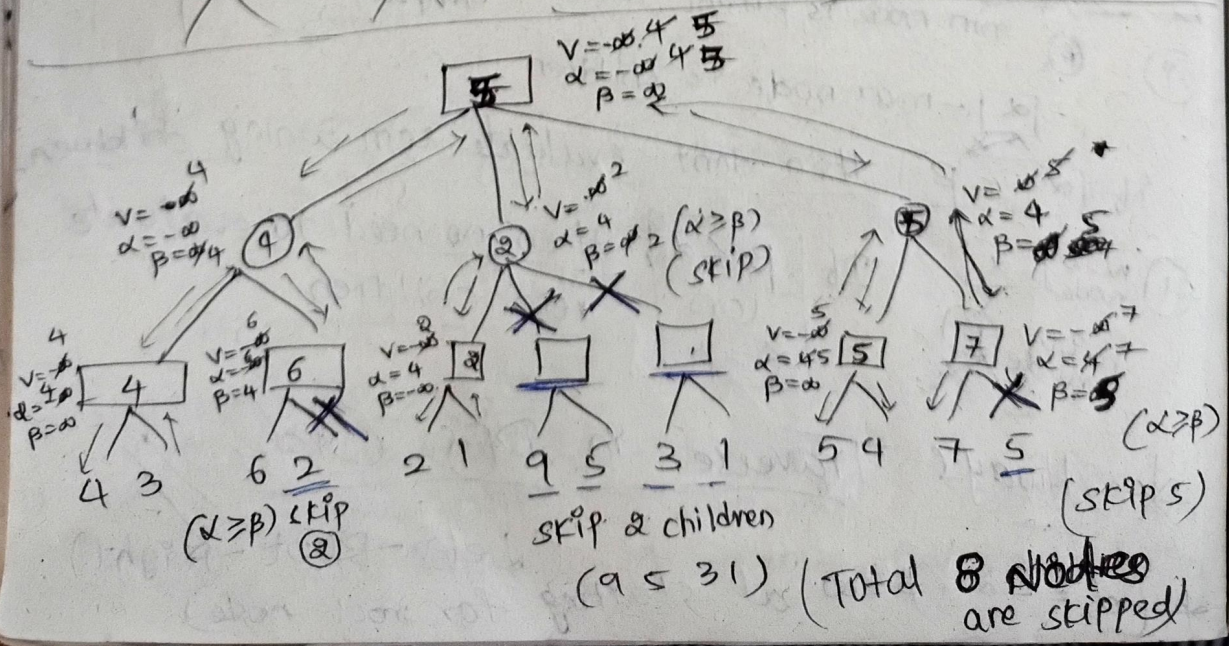
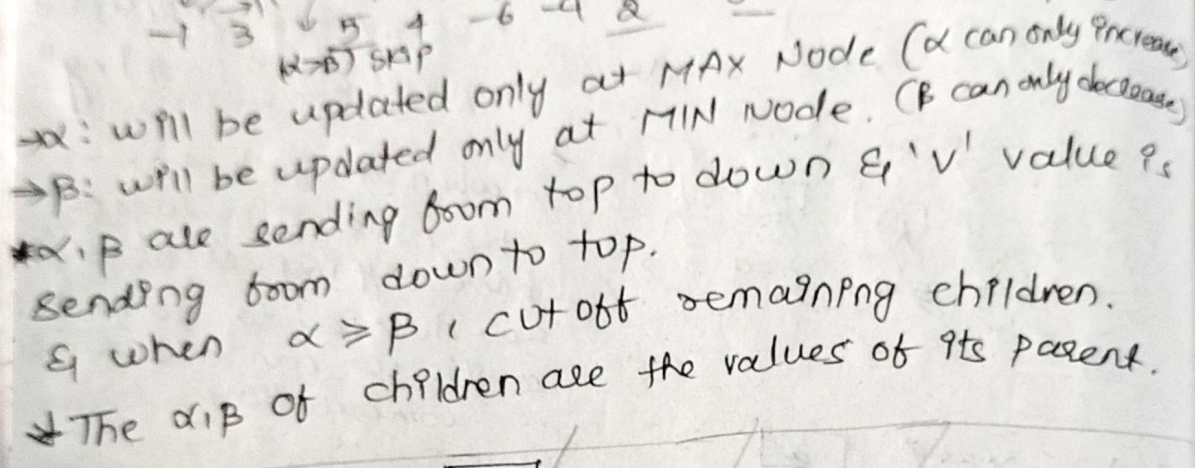
min node is parent.

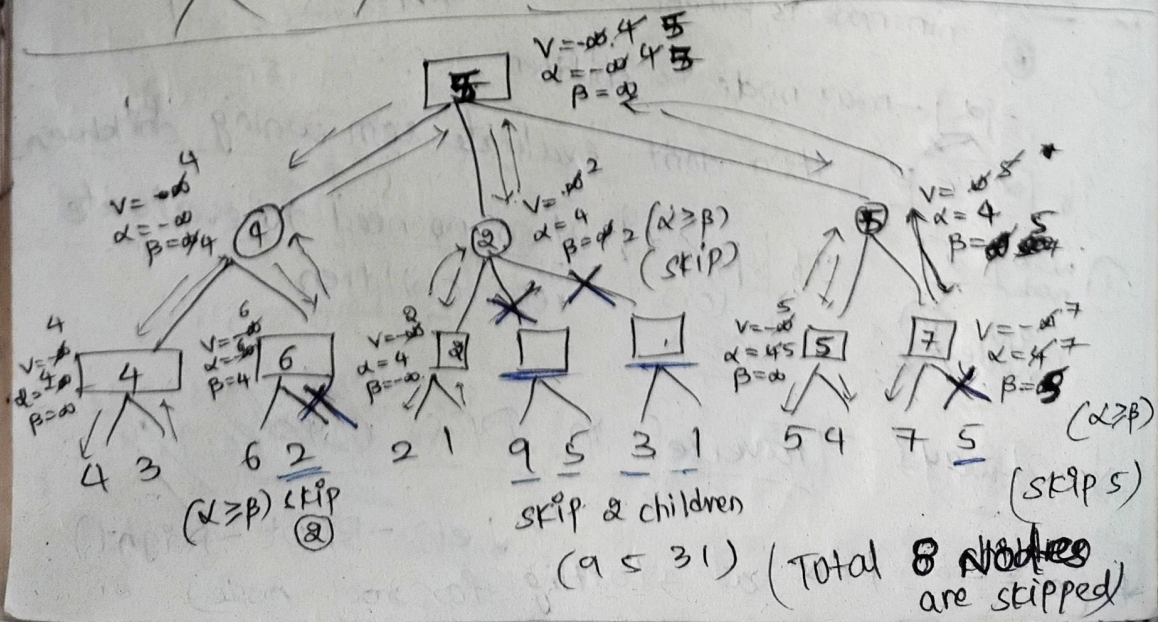
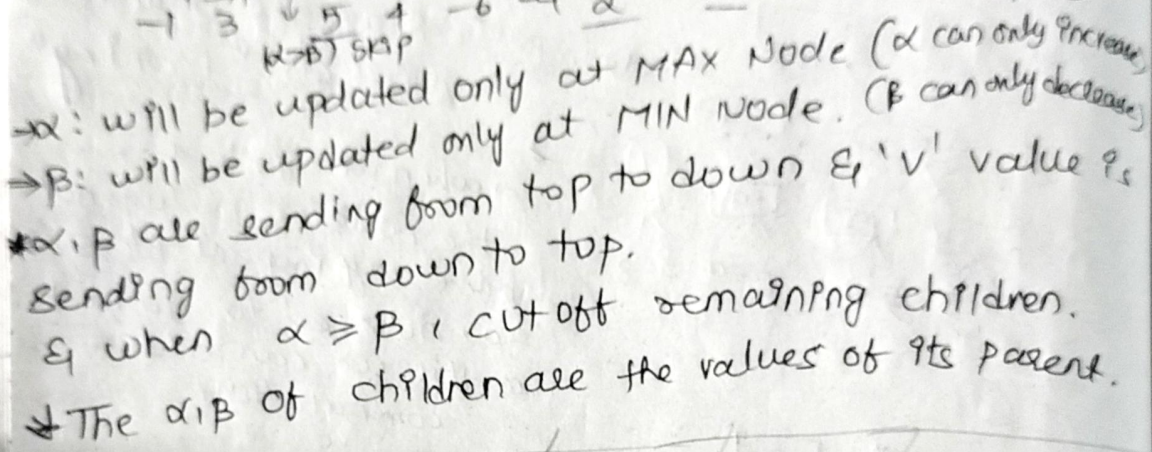
① ② - max node. 9 is children

if $\alpha \geq \beta$ then don't evaluate remaining children

② If $\beta \leq \alpha$ (or) $\alpha \geq \beta$ then no need to evaluate remaining children

* Always Traverse in DFS Order (Left-Root-Right) $\alpha = -\infty, \beta = \infty$ at starting for root node





Alpha-Beta pruning: $\alpha = -\infty, \beta = \infty$

Alpha-Beta (N, α, β):

if N is a terminal node:
return evaluation(N)

if N is a MAX node:
 $V \leftarrow -\infty$

for child c in MoveGen(N):

$V \leftarrow \max(V, \text{Alpha-Beta}(c, \alpha, \beta))$

$\alpha \leftarrow \max(\alpha, V)$

if $\alpha \geq \beta$: break.

else if N is a MIN node:

$V \leftarrow \infty$

for child c in MoveGen(N):

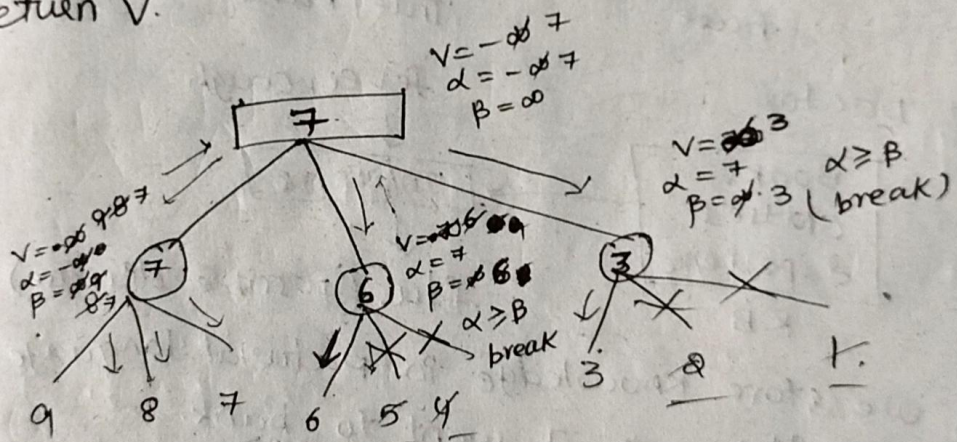
$V \leftarrow \min(V, \text{Alpha-Beta}(c, \alpha, \beta))$

$\beta \leftarrow \min(\beta, V)$

if $\alpha \geq \beta$: break.

return V .

Ex 7



Total 4 nodes are skipped. //

* ~~Knowledge & Reasoning~~

- Agent $\Rightarrow$ Intelligent program.

UNIT-I & UNIT-II - Search

- state space search
- Game Tree Search

UNIT-III, UNIT-IV - Logic based agent

UNIT-V - probability (Probabilistic Reasoning/Inferences).

Knowledge & Reasoning (before 1990)

type.

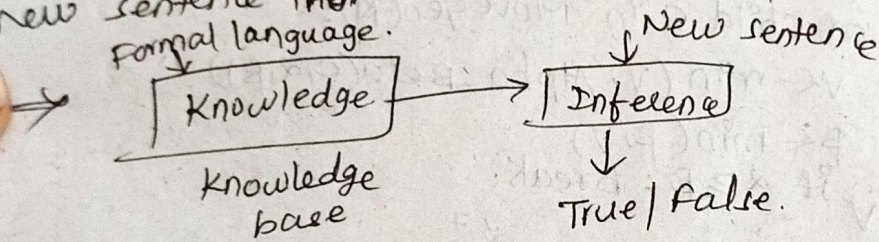
- propositional Logic
- predicate Logic (First order logic)
- Inference Algorithm/Rule.

→ These are languages & c/c "Formal Language"

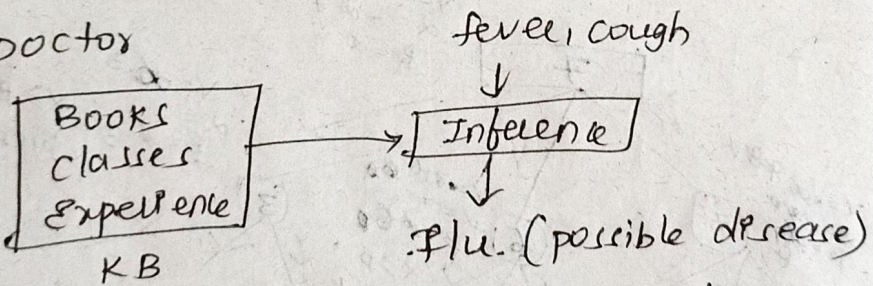
- Natural Language.

↓ programming lang are formal b/c they follow strict syntax.

* The storage of Knowledge is c/c "Knowledge base" & apply some Inference algo^m ^{which gives a new sentence then it decide it is true/false.} on that & if true a new sentence then it decide it is true/false.



Ex:1 Doctor



* If we store knowledge in natural language: →

1. Ambiguity. ⇒ I went to bank

CBI River bank.

That's why we use formal language.

→ which is not Ambiguous & also we can apply Inference algorithm.

→ Formal Language: →

Ex: → KB. All men are mortal
Socrates is a man → $a \rightarrow b$

↓

$\therefore$ Socrates is a mortal (true) $\underline{a \rightarrow c}$.

$b \rightarrow c$
↓
 $a \rightarrow c$

Ex 13
 KB $\boxed{\begin{array}{l} \text{Nothing is better than world peace} \\ \text{A burger is better than nothing} \end{array}}$ $b \rightarrow c$
 $a \rightarrow b$
 $\downarrow$
 $a = \text{A burger is better than world peace}$ (true) $\frac{a \rightarrow b}{a \rightarrow c}$

Ex 14
 KB $\boxed{\text{The barber shaves only them who do not shave themselves}}$
 $\downarrow$
 $a = \text{The barber shaves himself}$ (False)
 $\downarrow$
 (Russell's paradox) (we can't decide T/F)

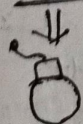
Ex 15
 KB $\boxed{\begin{array}{l} p \rightarrow q \\ p \end{array}}$ $\boxed{\begin{array}{l} \text{Rain} \rightarrow \text{Ground is wet} \\ \text{Rain} \end{array}}$
 $\downarrow$
 $a = q$
 $\downarrow$
 Inference
 Knowledge representn $\downarrow$
 Ex 16 $\boxed{\begin{array}{l} \text{Rain} \rightarrow \text{Ground is wet} \\ \text{Ground is wet} \end{array}}$
 $a = \text{Rain}$ (false)

* Our objective is to writing a knowledge in knowledge base in formal language. & apply inference on them.

* Diff types of Logic.

1. propositional logic } 0/1. (T/F)
2. first order logic }
3. Second order logic
4. Temporal logic $\rightarrow$ allow to infer abt time
5. Epistemic logic $\rightarrow$ multi agent logic.
6. Fuzzy logic $\rightarrow$ The sentence logic is b/w 0 & 1.
 $\rightarrow$ value b/w [0 to 1]

* Mine sweeper



$\rightarrow$ bomb put under the ground. & if we stand on that we will die.

Nothing is better than world peace. $b \rightarrow c$
 A burger is better than nothing. $a \rightarrow b$
 $\downarrow$
 $a =$ A burger is better than world peace. $\frac{a \rightarrow b}{a \rightarrow c}$
 (true)

The barber shaves only them who do not shave themselves.
 $\downarrow$
 $a =$ The barber shaves himself (false).
 $\downarrow$
 (Russell's paradox) (we can't decide T/F)

KB $\left[\begin{array}{l} p \rightarrow q \\ p \end{array} \right]$ $\left[\begin{array}{l} \text{Rain} \rightarrow \text{Ground is wet} \\ \text{Rain} \end{array} \right]$
 $\downarrow$ Inference $\alpha = q$ $\downarrow$ Knowledge representn $\alpha = \text{Ground is wet (true)}$
 Ex: $\left[\begin{array}{l} \text{Rain} \rightarrow \text{Ground is wet} \\ \text{Ground is wet} \end{array} \right]$
 $\alpha = \text{Rain (false)}$

* Our objective is to writing a knowledge in knowledge base in formal language. & apply inference on them.

* Diff types of Logic:

1. propositional logic } 0/1. (T/F)
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4. Temporal logic $\rightarrow$ allows to infer abt time
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6. Fuzzy logic $\rightarrow$ The sentence logic is b/w 0 & 1.
 $\rightarrow$ value b/w [0 to 1]

* Mine sweeper



$\rightarrow$ bomb put under the ground. & if we stand on that we will die.

	0	1	2	3	4
0	✓	✓	0	✓	0
1	✓		2	3	1
2	0	2	1	✓ 1	0
3	0	1	3	2	1
4	0	0	0	1	0

↑
 we need to cross & below the field we have certain mines. we have mine sweeper it tells how many mines are present around the field.
 → If it is 0 ⇒ No mines in surrounding.
 If it is 1 ⇒ There are mines in surrounding.
 Here we don't have search, before stepping we have to decide if it is safe/not. by using Inference.

Logic

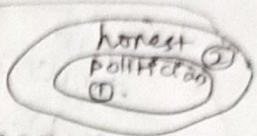
- ① Propositional Logic
 - ② First order Logic
- Language
 ↓
 sentences / expression
 ↓
 Truth values
 $\{0, 1\}, \{F, T\}, \{1, 0\}, \{\text{false}, \text{true}\}$

* Entailment

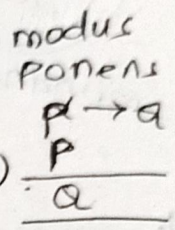
* $KB \models \alpha$ (Knowledge base entails α)
 $\Rightarrow \alpha$ is necessarily true when KB is true. Inference

Ex: KB: All politicians are honest
Sambit is a politician True. $\left[\begin{array}{l} p \rightarrow q \\ p \\ \hline \therefore q \end{array} \right]$
 α : Sambit is honest $\Rightarrow$ necessarily True
 $\therefore \boxed{KB \models \alpha} //$

Ex. 1.7
 KB: All politicians are honest
 gambit is honest
 α : gambit is politician. (not necessarily true)
 $\boxed{KB \not\models \alpha} //$



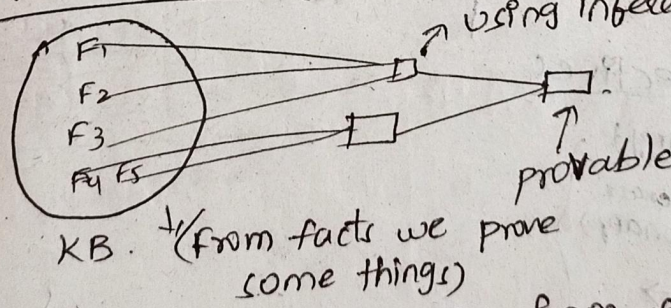
Ex. 2
 KB: If it rains ground will get wet
 It has rained.
 α : Ground is wet (necessarily true)
 $\boxed{KB \models \alpha} //$



Ex. 3
 KB: If it rains ground will get wet
 Ground is wet
 α : It has rained. (not necessarily true)
 $\boxed{KB \not\models \alpha} //$

Ex. 4
 KB: If you study well, you will get good marks
 you get good marks.
 α : You have studied well. (not necessarily true)
 $\boxed{KB \not\models \alpha} //$

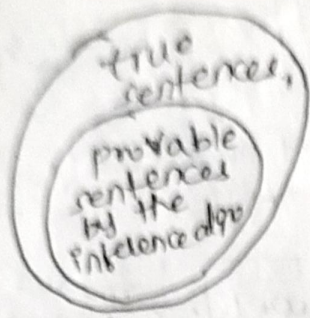
* Provability - $(KB \vdash \alpha)$



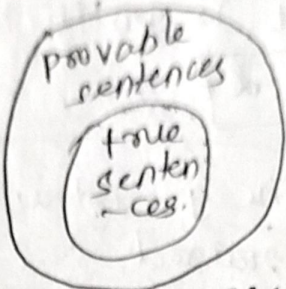
KB. $\downarrow$ (from facts we prove some things)

- * deriving new sentence from existing sentences in KB using rules of inference.
- * $\boxed{KB \vdash \alpha}$ (KB derives α)

- * Soundness & completeness of Inference Algorithm
- * From entailment & provability we generated soundness & completeness.



→ This doesn't infer wrong sentences but doesn't cover all true sentences

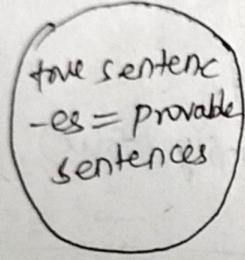


(delusional) we speak all truth & also additional info which is not true.

soundness.

completeness.

* Inference algo should be both sound & complete.



→ Inference Algo

* This can prove all true sentences & it won't produce any wrong sentences.

* sentences

* It is of 2 types:

① Atomic/primitive

— simplest & indivisible.

P: It is raining. (T/F)

Q: Ground is wet (T/F)

② compound sentences

* combining ~~some~~ sentences using logical connectives (and, or)

Ex: It is raining and ground is wet.

* Logical connectives

AND: $\wedge$ (Binary)

OR: $\vee$

NOT: $\neg$ → (unary)

Implication: $\rightarrow$

By conditional: $\leftrightarrow$

XOR: $\oplus$

XNOR: $\odot$

equivalence: $\equiv$

Syntax & Semantics

Syntax $\Rightarrow$ structure/form

$\neg p \wedge q$ is well formed formula.

$p \wedge q$ is well formed formula.

$p \wedge$ is not well formed formula.

Semantics $\Rightarrow$ meaning.

p : It is raining.

$\neg p$: It is not raining.

p	$\neg p$
T	F
F	T

Ex: $p \rightarrow q$

p is raining $\rightarrow$ q is wet

T	T
F	T
T	F
F	F

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$p \leftrightarrow q$
T	T	T
F	T	F
T	F	F
F	F	T

$\leftrightarrow$ is only if (equivalent operator)

You will get attendance if & only if you present in the class.

Attendance	$\leftrightarrow$	present in class
T	T	$\rightarrow$ T
T	F	$\rightarrow$ F
F	T	$\rightarrow$ F
F	F	$\rightarrow$ T

Sentences $\Rightarrow$

$$(\alpha \rightarrow \beta) \equiv (\neg \alpha \vee \beta)$$

$\downarrow$
(equiv operator)

α	β	$\alpha \rightarrow \beta$	$\neg \alpha$	$\neg \alpha \vee \beta$	$\alpha \leftrightarrow \beta$
T	T	T	F	T	T
T	F	F	F	F	F
F	T	T	T	T	F
F	F	T	T	T	F

Tautology

* entailment understanding with truth table

$\alpha \rightarrow \beta, \alpha \models \beta$

$\alpha \rightarrow \beta$ is true & then β is necessarily true if α is true.

α	β	$\alpha \rightarrow \beta$
T	T	T
T	F	F
F	T	T
F	F	T

$\alpha \rightarrow \beta$ is True

α	β	$\alpha \rightarrow \beta$
T	T	T
F	T	T
F	F	T

α is true

α	β	$\alpha \rightarrow \beta$
T	T	T

$\therefore \beta$ is true

Syntax & Semantics

Syntax $\Rightarrow$ structure/form
 well formed formula.
 $P \wedge Q$: well formed formula.
 $P \wedge$: not well formed formula.

Semantic $\Rightarrow$ meaning.

P : It is raining.
 $\neg P$: It is not raining.

P	$\neg P$
T	F
F	T

Ex: $P \rightarrow Q$
 It is raining $\rightarrow$ Grd is wet

T	T
F	T
T	F
F	F

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Ex: $P \leftrightarrow Q$

P	Q	$P \leftrightarrow Q$
T	T	T
F	T	F
T	F	F
F	F	T

$\leftrightarrow$ & only if (equivalent operator)
 You will get attendance if & only if you present in the class.
 Attendance $\leftrightarrow$ present in class

T	T	$\rightarrow$ T
T	F	$\rightarrow$ F
F	T	$\rightarrow$ F
F	F	$\rightarrow$ T

Sentences

$(\alpha \rightarrow \beta) \equiv (\neg \alpha \vee \beta)$
 $\downarrow$
 (equiv operator)

α	β	$\alpha \rightarrow \beta$	$\neg \alpha$	$\neg \alpha \vee \beta$	$\alpha \leftrightarrow \beta$
T	T	T	F	T	T
T	F	F	F	F	F
F	T	T	T	T	T
F	F	T	T	T	F

$\downarrow$
 Tautology.

* entailment understanding with truth table

$\alpha \rightarrow \beta$, $\alpha \models \beta$.
 $\alpha \rightarrow \beta$ is true & then β is necessarily true
 α is true.

α	β	$\alpha \rightarrow \beta$
T	T	T
T	F	F
F	T	T
F	F	T

$\alpha \rightarrow \beta$ is True

α	β	$\alpha \rightarrow \beta$
T	T	T
F	T	T
F	F	T

α is true

α	β	$\alpha \rightarrow \beta$
T	T	T

$\therefore \beta$ is true

ii) $\alpha \rightarrow \beta, \beta \vdash \alpha$

α	β	$\alpha \rightarrow \beta$
T	T	T
T	F	F
F	T	T
F	F	T

$\alpha \rightarrow \beta$ is true

α	β	$\alpha \rightarrow \beta$
T	T	T
F	T	T
F	F	T
T	F	F

β is true

α	β	$\alpha \rightarrow \beta$
T	T	T
F	T	T
F	F	T
T	F	F

α is not necessarily true

$\alpha \rightarrow \beta$ is true & β is true, then α is not necessarily true

$\therefore \alpha \rightarrow \beta, \beta \not\vdash \alpha$

* Sound/Valid Rule of Inference

* How do we know a rule is sound/valid?

* A rule $\{\alpha, \beta\} \vdash \delta$ (delta), is valid if δ is tautology.

9) $\boxed{\alpha \rightarrow \beta, \alpha \vdash \beta}$ ($\alpha \rightarrow \beta$ & α are true then α derives β)

α	β	$\alpha \rightarrow \beta$	$(\alpha \rightarrow \beta) \wedge \alpha$	$((\alpha \rightarrow \beta) \wedge \alpha) \rightarrow \beta$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Tautology

ii) $\boxed{\alpha \rightarrow \beta, \neg \beta \vdash \neg \alpha}$

α	β	$\alpha \rightarrow \beta$	$(\alpha \rightarrow \beta) \wedge \neg \beta$	$((\alpha \rightarrow \beta) \wedge \neg \beta) \rightarrow \neg \alpha$	$\neg \alpha$
T	T	T	F	T	F
F	T	T	F	T	F
T	F	F	F	T	T
F	F	T	F	T	T

Tautology

$\therefore \alpha \rightarrow \beta$ is true & $\neg \beta$ is true then $\neg \beta \vdash \neg \alpha$ is valid

$\therefore$ This inference is valid //

e.g. $\boxed{\text{If it rains, ground is wet.}} \quad (\alpha \rightarrow \beta)$
 $\text{Ground is not wet} \quad (\neg \beta)$

$\Downarrow$
 It is not raining. $\rightarrow (\neg \alpha)$ is derived & valid.

2v2, 8-8-1 pvd.

[illegible]

(Tautology)

$$((\alpha \rightarrow \beta) \wedge (\alpha \vee \gamma) \wedge (\gamma \rightarrow \delta)) \rightarrow (\beta \vee \delta)$$

⇒ Tautology

$$\alpha \rightarrow \beta, \quad \beta \rightarrow \delta \vdash \alpha \vee \delta$$

valid, rule of
Inference.

$\alpha \rightarrow \beta, \alpha \vee \gamma, \gamma \rightarrow \delta$ are true then $(\gamma \rightarrow \delta) \vdash \text{PVS } \gamma$ true //

* $\{ \alpha, \beta \} \vdash \gamma$ is valid if $(\alpha \wedge \beta) \rightarrow \gamma$ is a tautology

① Modus Tollens

$$\frac{\alpha \rightarrow \beta}{\neg \beta} \quad \frac{\neg \beta}{\neg \alpha}$$

(2) conjunction

$$\begin{array}{c} \alpha \\ \beta \\ \hline \alpha \wedge \beta \end{array}$$

(3) Addition.

$$\frac{\alpha}{\alpha \vee \beta}$$

④ simplification.

$$\frac{\alpha \wedge \beta}{\alpha}$$

⑤ Disjunctive syllogism.

$$\frac{\alpha \vee \beta}{\beta}$$

Ex: I will go to class or I will stay in hostel.
I didn't go to class
I stayed in hostel

⑥ Hypothetical syllogism.

$$\frac{\alpha \rightarrow \beta}{\beta \rightarrow \gamma}$$

⑦ constructive Dilemma.

$$\frac{(\alpha \rightarrow \beta) \wedge (\gamma \rightarrow \delta)}{\alpha \vee \gamma}$$

Ex: If I study hard I will pass the exam and
If I join the football team I will make new friends.
I studied hard or I joined the football team.
I passed the exam or I made new friends.

⑧ Destructive Dilemma

$$\frac{(\alpha \rightarrow \beta) \wedge (\gamma \rightarrow \delta)}{\neg \beta \vee \neg \delta}$$

* Rules of substitution

1) $\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta \rightarrow$ tautology.

α	β	$\neg \alpha$	$\alpha \rightarrow \beta$	$\neg \alpha \vee \beta$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Common Rules

① $\alpha \vee T \equiv T$

$$\alpha \wedge T \equiv T$$

$$\alpha \vee F \equiv \alpha$$

$$\alpha \wedge F \equiv F \rightarrow \text{contradiction}$$

$$\alpha \wedge \neg \alpha \equiv F$$

$$\alpha \vee \neg \alpha \equiv T$$

$$\neg(\neg \alpha) = \alpha$$

$$\alpha \vee (\beta \vee \gamma) \equiv (\alpha \vee \beta) \vee \gamma \rightarrow \text{associative}$$

$$\alpha \wedge (\beta \wedge \gamma) \equiv (\alpha \wedge \beta) \wedge \gamma$$

$$\alpha \vee \beta \equiv \beta \vee \alpha \quad \alpha \wedge \beta \equiv \beta \wedge \alpha \quad \text{commutative}$$

$$\neg(\alpha \vee \beta) = \neg \alpha \wedge \neg \beta \quad \neg(\alpha \wedge \beta) = \neg \alpha \vee \neg \beta \quad \text{Demorgan's law}$$

$$\alpha \wedge (\beta \vee \gamma) = (\alpha \wedge \beta) \vee (\alpha \wedge \gamma) \quad \alpha \vee (\beta \wedge \gamma) = (\alpha \vee \beta) \wedge (\alpha \vee \gamma) \quad \text{distribution}$$

$$\alpha \rightarrow \beta \equiv \neg \alpha \vee \beta \equiv \neg \beta \rightarrow \neg \alpha \quad (\text{contrapositive})$$

Rain $\rightarrow$ Ground wet
Ground not wet $\rightarrow$ No Rain.

$$(\alpha \leftrightarrow \beta) \equiv (\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha) \rightarrow \text{equivalence}$$

$$((\alpha \rightarrow \beta) \wedge (\alpha \rightarrow \neg \beta)) \equiv \neg \alpha \rightarrow \text{Absurdity}$$

α	β	$\neg \beta$	$\alpha \rightarrow \beta$	$\alpha \rightarrow \neg \beta$	$(\alpha \rightarrow \beta) \wedge (\alpha \rightarrow \neg \beta)$
T	T	F	T	F	F
T	F	T	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

* Propositional Logic.

simplest logical language.

Atomic sentences as symbol.

i.e. $P =$ Earth is flat (T/F)

$Q =$ Tomato is fruit (T/F)

} we can't divide it further.

→ Domain: set of sentences.

$$IP = \{A \mid B \dots P_n\}$$

→ Logical connectives: $\vee, \wedge, \neg$.

→ Brackets: $(), \{ \}$

→ Constants: T, F .

* Inference

- ① Truth Table $\Rightarrow$ (exponential)
- ② Natural deduction. (inference rules).
- ③ Resolution Refutation (proof by contradiction)
- ④ Horn clause Inference

② Natural Deduction (rules of Inference)

Ex: Alice like Math & she likes stories.

① If she likes math then she like algebra.
If she likes algebra and like physics, she will go to college.

she does not like stories or she likes physics.
she doesn't like chemistry and history.

prove: Alice will go to college

KB: Alice like math
a: she like stories.
s: she like algebra
p: she like physics.

u: she like chemistry

v: she like history.

w: she go to college.

KB:

i) $p \wedge a$

ii) $p \rightarrow s$ iii) $(s \wedge s) \rightarrow w$

iv) $(\neg a \vee s)$

v)

~~$u \wedge v$~~ : ~~$(u \rightarrow v)$~~
 ~~$\neg u \wedge \neg v$~~ : ~~$(u \rightarrow \neg v)$~~

Goal: $\vdash$

Proof:

- | | |
|----------------------------------|------------------------------|
| 1. $P \wedge Q$ | premise |
| 2. $P \rightarrow R$ | premise |
| 3. $(R \wedge S) \rightarrow T$ | premise |
| 4. $\neg Q \vee S$ | premise |
| 5. P | simplification (1) |
| 6. Q | simplification (1) |
| 7. R | (2, 5) Modus ponens |
| 8. S | (4, 6) Disjunctive syllogism |
| 9. $R \wedge S$ | (7, 8) conj ⁿ |
| 10. <u>T</u> (Goal) | (9, 3) Modus ponens |

If it is Saturday today, then we play soccer or basketball.

If soccer field is occupied then we don't play soccer.

It is Saturday today and soccer field is occupied.

prove: We play basketball or volleyball.

- Solⁿ:
- P : It is Saturday today
 - Q : we play soccer
 - R : we play basketball
 - S : soccer field is occupied
 - T : we play volleyball.

i) $P \rightarrow (Q \vee R)$ ii) $S \rightarrow \neg Q$ iii) $P \wedge S$ iv) T

Goal: $\vdash R \vee T$

Solⁿ:

- | | |
|-------------------------------|---------------------------|
| i) $P \rightarrow (Q \vee R)$ | premise |
| ii) $S \rightarrow \neg Q$ | premise |
| iii) $P \wedge S$ | premise |
| iv) P | iii, simplif ⁿ |

- iv) $\exists$ iii, simplifn
 v) $\neg$ (iv, ii) MP.
 vi) $\forall$ (i, iv) MP.
 vii) R (vi, v) Disjunctive syllogism.
 viii) T premise
 ix) $\frac{RVT}{(Goal)}$ (vii, viii) Addition.

* conjunctive Normal form (CNF). [POS]

$$SOP: (4 \times 5) + (6 \times 7)$$

$$POS: (4+5) * (6+7)$$

$\Rightarrow$ It is product of sums.

* It is conjunction of clauses (AND).

* Each clause is disjunction (OR) of literals.

* A literal is an atomic sentence (ex: p) or its negation ($\neg p$).

$$\text{Ex: } (p \vee \neg q) \wedge (r \vee s) \wedge (\neg p \vee q \vee t) \rightarrow \text{CNF.}$$

Rules for conversion of CNF ($\wedge, \vee, \neg$).

1. Eliminate other connectives like $\rightarrow, \leftrightarrow, \oplus$

$$\begin{aligned}
 p \leftrightarrow q &\equiv (p \rightarrow q) \wedge (q \rightarrow p) \\
 &\equiv (\neg p \vee q) \wedge (\neg q \vee p)
 \end{aligned}$$

2. push negation inwards.

$$\text{Ex: } \neg(p \wedge q) \equiv \neg p \vee \neg q.$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\neg(\neg p) \equiv p.$$

3. Distribute $\vee$ over $\wedge$.

$$\text{Ex: } p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

4. Simplify / Remove duplicates.

$$\text{Ex: } p \vee p \equiv p.$$

$$(P \vee Q) \rightarrow R$$

$$\equiv (\neg(P \vee Q) \vee R)$$

$$\equiv (\neg P \wedge \neg Q) \vee R$$

$$\equiv R \vee (\neg P \wedge \neg Q)$$

$$\equiv (R \vee \neg P) \wedge (R \vee \neg Q) \Rightarrow \text{CNF}$$

$$\neg(P \vee (Q \wedge R))$$

$$\equiv \neg P \wedge \neg(Q \wedge R)$$

$$\equiv \neg P \wedge (\neg Q \vee \neg R) \Rightarrow \text{CNF}$$

$$(P \leftrightarrow Q) \vee R$$

$$\equiv (P \rightarrow Q) \wedge (Q \rightarrow P) \vee R$$

$$\equiv ((\neg P \vee Q) \wedge (\neg Q \vee P)) \vee R$$

$$\equiv R \vee ((\neg P \vee Q) \wedge (\neg Q \vee P))$$

$$\equiv (R \vee \neg P \vee Q) \wedge (R \vee \neg Q \vee P) \equiv \text{CNF} //$$

③ Resolution Refutation

* To prove $\{\alpha, \beta\} \vdash \gamma$, we can show: $\neg$
 $(\alpha \wedge \beta) \rightarrow \gamma$ is tautology.

$$\text{Ex: } \alpha, \alpha \rightarrow \beta \vdash \beta.$$

$$\alpha \wedge (\alpha \rightarrow \beta) \wedge \neg \beta.$$

α	β	$\neg \beta$	$\alpha \rightarrow \beta$	$\alpha \wedge (\alpha \rightarrow \beta) \wedge \neg \beta$
T	T	F	T	F
T	F	T	F	F
F	T	F	T	F
F	F	T	T	F

contradiction.

we can show $(\alpha \wedge \beta \wedge \neg \gamma)$ is contradiction / always false.

- To prove $\{ \alpha, \beta \vdash \gamma \}$
- we can show $(\alpha \wedge \beta) \rightarrow \gamma$ is tautology
- we can show $(\alpha \wedge \beta \wedge \neg \gamma)$ is contradiction / always false
- $KB \vdash \phi$ is true if and only if $KB \wedge \neg \phi$ is contradiction

(KB = set of ~~sentences~~ statements)

1. convert KB to CNF form

2. Repeatedly apply:

Resolution Rule: $\frac{[A \vee P] \quad [\neg P \vee B]}{A \vee B}$

Stop if you reach false. ~~no~~ no more clauses can be combined.

Ex: 1. A

2. $A \rightarrow B$

3. $B \rightarrow C$

prove: C.

Sol: convert to CNF:

1. A

2. $\neg A \vee B$

3. $\neg B \vee C$

proof: C.

$A \wedge (\neg A \vee B) \wedge (\neg B \vee C) \wedge \neg C$ is contradiction (we need to prove)

[A] [$\neg A \vee B$] [$\neg B \vee C$] [$\neg C$]

[B]

[C]

[false] (empty clause = false)
→ stop.

return true.

i.e It is provable from KB.

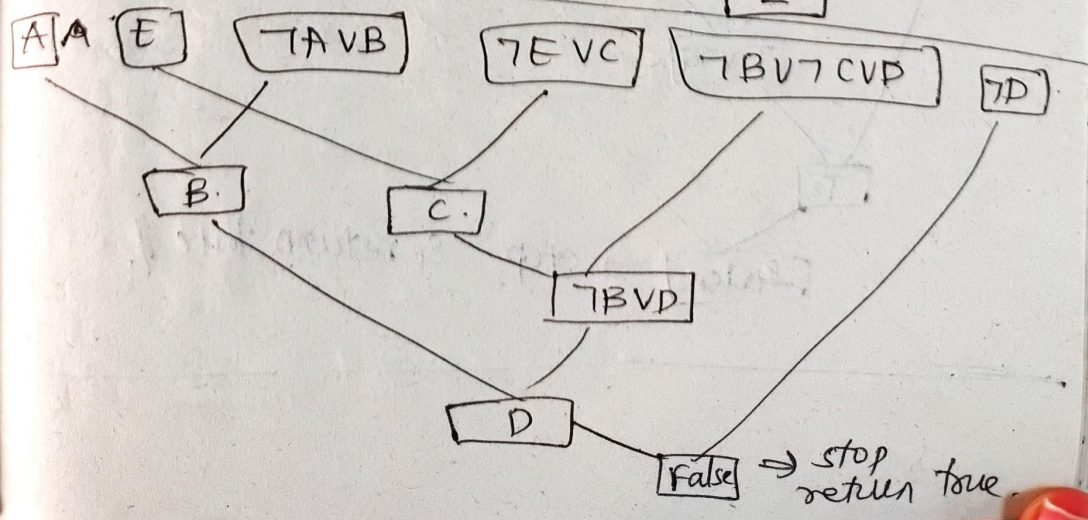
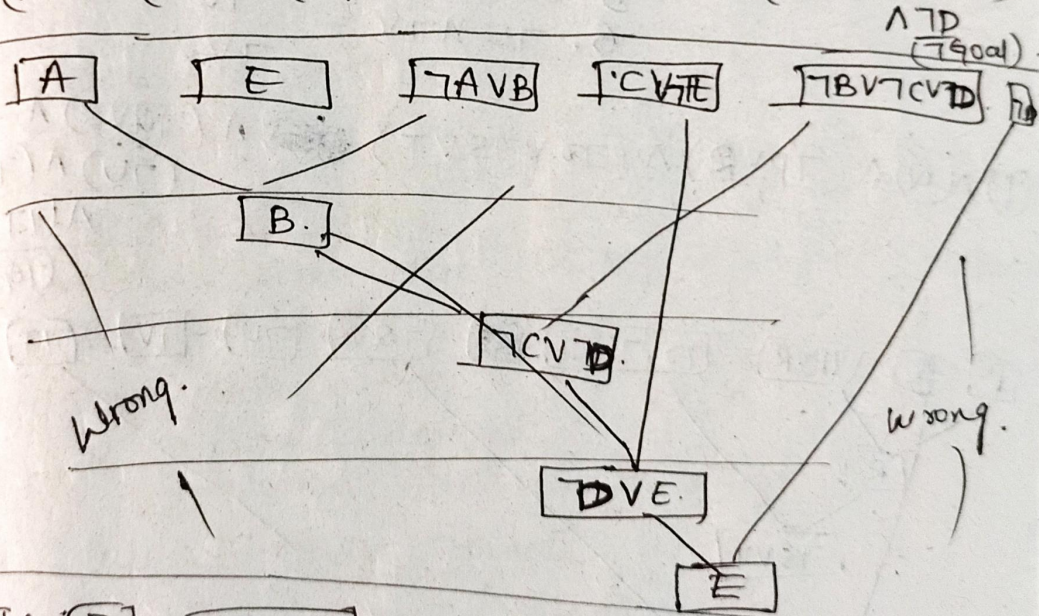
KB: A
 E
 $A \rightarrow B$
 $E \rightarrow C$
 $B \wedge C \rightarrow D$

prove: D.

convert to cnf

1. A
2. E
3. $\neg A \vee B$
4. $\neg E \vee C$
5. $\neg(B \wedge C) \vee D \equiv \neg B \vee \neg C \vee D$

1. $(A) \wedge (E) \wedge (\neg A \vee B) \wedge (\neg E \vee C) \wedge (\neg B \vee \neg C \vee D)$

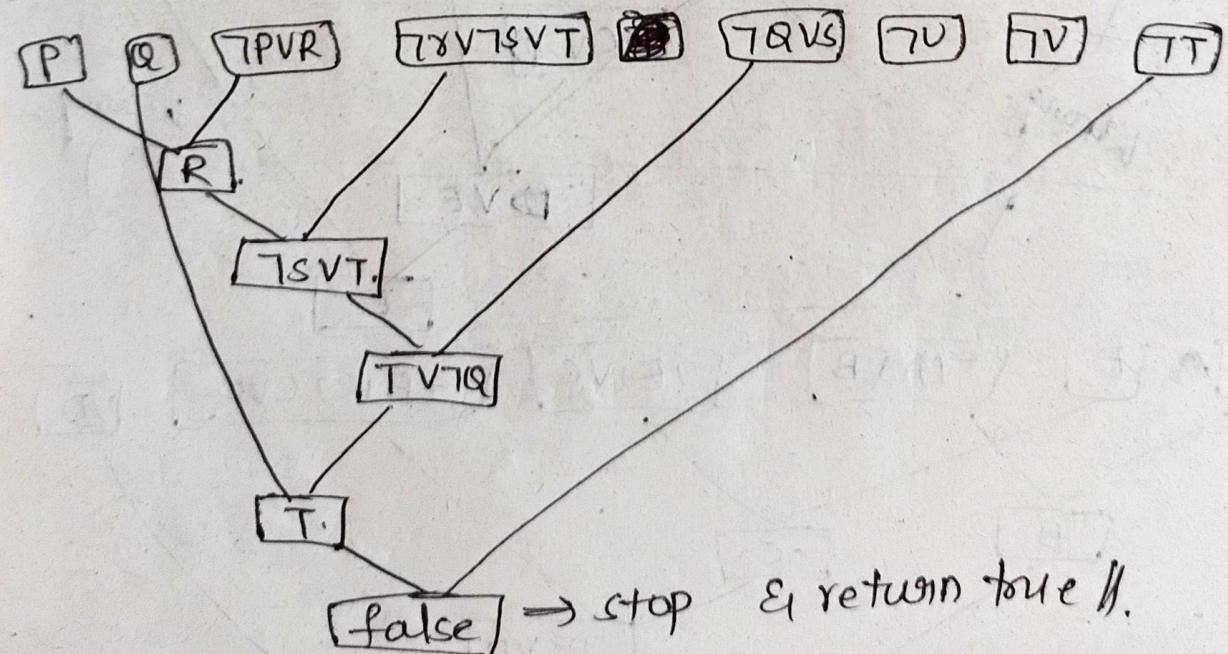


(5) KB: $P \wedge Q$.
 $P \rightarrow \delta$
 $(\delta \wedge S) \rightarrow T$
 δ .
 $\neg Q \vee S$.
 $\neg U \wedge \neg V$.
prove: T.

[Alice problem].

- soln: 1. $P \wedge Q \Rightarrow$ 1. P
 2. Q
 3. $\neg P \vee R$.
 4. $\neg(\delta \wedge S) \vee T \equiv \neg \delta \vee \neg S \vee T$
~~5. P.~~
 5. $\neg Q \vee S$
 6. $\neg U \wedge \neg V \equiv \neg U$
 $\neg V$

$$(P) \wedge (Q) \wedge (\neg P \vee R) \wedge (\neg \delta \vee \neg S \vee T) \wedge (\neg Q \vee S) \wedge (\neg U) \wedge (\neg V) \wedge (TT) \wedge (TT)$$



false $\Rightarrow$ stop & return true //

* Horn clause Inference.
 * A Horn clause is a clause where at most one literal is positive. (0, or 1 +ve literal)
 ex: $\neg P \vee Q$. ✓
 $\neg P \vee \neg Q \vee R$. ✓ ($P \wedge Q \rightarrow R$)

$\neg P \vee Q \vee R$. X.

$\neg P \vee \neg Q$. ✓

* forms of Horn clause $\Rightarrow$

1. Definite clause / Rule $\Rightarrow$.
 Has exactly one +ve literal.

ex: $\neg P \vee \neg Q \vee R$.

2. Fact $\Rightarrow$ Just a single literal

ex: P , $\neg P$.

3. Goal clause: No positive literals.

ex: $\neg P \vee \neg Q$.

* Forward chaining

$\rightarrow$ start from known facts in KB and apply rules to infer new facts.

$\rightarrow$ continue until either

— Goal is derived

— No more facts can be added.

* It is also ds "Data Driven"

ex: KB: $P \rightarrow Q$. — premise

$L \wedge M \rightarrow P$. — conclusion.

$B \wedge L \rightarrow M$.

$A \wedge P \rightarrow L$

$A \wedge B \rightarrow L$

A

B

prove: Q

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Solve CNF

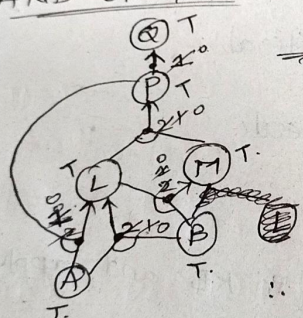
1. $\neg P \vee Q$
2. $\neg L \vee \neg M \vee P$
3. $\neg B \vee \neg L \vee M$
4. $\neg A \vee \neg P \vee L$
5. $\neg A \vee \neg B \vee L$
6. A
7. B

everything is in horn clause (rules)

facts

proof: Q

* AND-OR Graph.



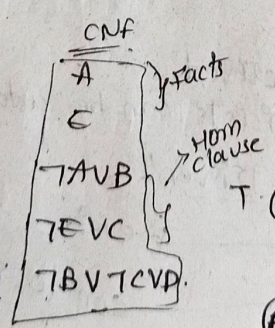
we start with facts & with that what is next fact coming true, add them & use it for next rule. Until our goal b/m true.

∴ At last Q is "True"

(A is true then it req 1)

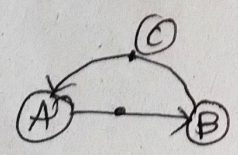
- Ex 1
- A
 - E
 - $A \rightarrow B$
 - $E \rightarrow C$
 - $B \wedge C \rightarrow D$

prove: D



∴ P is True

Ex 2



⇒ cycle, we need to avoid these cycles.

Implementation (Code)

UNIT-V - Probability

- Bayesian Network
- * Basics of probability
- Random variable
- Rain R ∈ {0, 1}
- sunshine S ∈ {0, 1}

S	R	P(S=R)
0	0	0.2
0	1	0.08
1	0	0.17
1	1	0.55

* Any prob dist

* In random probability, where is 0/1.

* Marginal

* There are

$P(S) =$

$P(S=0) =$

$\Rightarrow P(S=0) =$

$P(R=1) =$

* condition

$P(S|R=1) =$